

Monolithic versus Modular Group Surface Codes for a Quantum Adder: Construction Tradeoffs and Classwise Decoder Diagnostics

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(Dated: June 21, 2026)

Group surface codes (GSCs), introduced by Manjunath et al. [1], generalize the standard $\mathbb{Z}_2$ surface code to an arbitrary finite group, enabling non-Clifford logical operations through the group structure. We present a task-level case study of quantum arithmetic in the GSC framework, constructing two implementations of a reversible full quantum adder and analyzing the resulting monolithic-versus-modular tradeoff. Construction A encodes the adder into a single GSC whose group G_{FA} has order 2^{13} , achieving a leading-order space-time volume of $130d^3$ under a compact-square benchmark with fully transversal gates. Construction B takes a modular approach, using two G_{CCX} codes of order 64 combined with lattice surgery for CX gates, at a leading-order space-time volume of $250d^3$, but with substantially more tractable decoding. We develop a classwise minimum-weight perfect-matching decoder for Construction B, exploiting the self-inverse conjugacy classes and self-dual irreducible representations of G_{CCX} , and prove single-fault correctness for code distance $d \geq 5$. Numerical simulations under an effective class-chain noise model—which does not establish a fault-tolerance threshold—show decoder behavior beyond the single-fault regime in both code-capacity and phenomenological settings. A two-edge diagnostic under the exact group-valued face flux exposes a concrete limitation of the classwise approximation: in a uniform-combinatorial diagnostic, 28.1% of adjacent two-edge configurations lead to logical failure when decoded classwise. More broadly, the monolithic G_{C^nX} family incurs an edge-register qubit cost of $q = 2^n + n$, growing exponentially in the number of controls, which constrains the scalability of the monolithic approach to higher-arity gates.

I. INTRODUCTION

Achieving universal fault-tolerant quantum computation remains a central challenge in quantum information science. The standard approach combines the surface code [2] with magic state distillation [3–5] to supplement Clifford gates with non-Clifford operations necessary for universality. While recent advances in magic state cultivation have dramatically reduced the associated overhead [3], the resource cost remains substantial.

Group surface codes (GSCs), recently introduced by Manjunath et al. [1], offer an alternative route. By replacing the $\mathbb{Z}_2$ symmetry of the surface code with an arbitrary finite group G , the GSC framework enables non-Clifford logical gates, such as the Toffoli gate (CCX), to be realized through the group structure, potentially circumventing the need for magic state distillation entirely. Whether this potential translates into a practical advantage over magic state distillation at realistic error rates and circuit depths remains an open question that depends on decoder development, threshold analysis, and hardware-specific costs not addressed here.

In this work, we present a task-level case study: rather than pursuing general universality, we focus on a concrete computational primitive—the reversible full quantum adder—and analyze the construction and decoding tradeoffs that arise when implementing it within the GSC framework. Arithmetic circuits built from CCX and CX gates are fundamental primitives for algorithms such as Shor’s factoring algorithm and quantum simulation. Our goal is to realize this primitive within the GSC framework and analyze the resulting tradeoffs in implementation as

well as the decoder.

We present two constructions: Construction A, which encodes the full adder into a single GSC whose group G_{FA} is of order 2^{13} , generated by all the gates appearing in the adder circuit; and Construction B, which takes a modular approach, combining two copies of the G_{CCX} code (order 64) [1] with standard lattice surgery for the CX gates [6]. A quantitative comparison under a compact-square resource benchmark shows that Construction A has smaller leading-order space-time volume through fully transversal gates, while Construction B requires a decoder for a group 128 times smaller, making it substantially more tractable. We develop a classwise minimum-weight perfect-matching decoder for Construction B, prove single-fault correctness for $d \geq 5$, and present numerical simulations that test decoder performance in the multi-fault regime while identifying the boundary of the classwise approximation through a two-edge diagnostic.

II. BACKGROUND

A. Group Surface Codes

A group surface code is defined by a finite group G and a cellulation of a two-dimensional surface [1]. In the standard $\mathbb{Z}_2$ surface code, physical qubits live on edges of the lattice and stabilizers correspond to faces and vertices. In GSCs, this construction is generalized: the local Hilbert space on each edge is $\mathbb{C}[G]$, the group algebra of G with basis states $|g\rangle$ labeled by group elements g . The

code space is defined by stabilizer conditions associated with faces and vertices of the cellulation, analogous to the plaquette and vertex operators of the toric code [1]. Logical operations on the GSC are implemented by left multiplication operators, $L_g|h\rangle = |g \cdot h\rangle$ and right multiplication $R_g|h\rangle = |h \cdot g^{-1}\rangle$, preserving the code space and implementing logical operations on the encoded information [1]. A key feature of this framework is the *extension* and *splitting* operations, allowing transitions between surface codes associated with different groups, allowing a group to be enlarged to access non-Clifford gates, and then returning to a simpler code.

B. Reversible Full Quantum Adder

The quantum adder circuit that we analyze is given by a circuit on four qubits involving the five gates $CCX_{124}, CX_{12}, CCX_{234}, CX_{23}, CX_{12}$, which is described in [7], and pictured in Fig. 1. It takes three input bits (a, b, c_{in}) and produces the output bits ($a, b, s = a \oplus b \oplus c_{in}, c_{out} = ab \oplus c_{in}(a \oplus b)$).

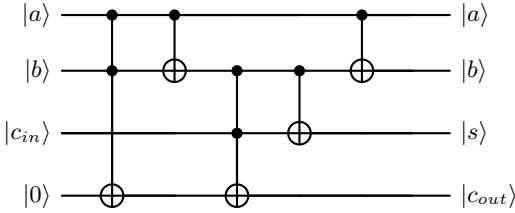


FIG. 1. Quantum Circuit for the Full Adder, where the input is four bits; the two being summed a, b , a carried bit c_{in} , and a zero bit. The output is the bits a, b , their sum s , and the carried bit c_{out}

III. FULL QUANTUM ADDER GSC CONSTRUCTIONS

A. Single GSC Implementation of the Full Adder

We can extend the G_{CCX} construction in [1] to find the GSC for the full adder circuit. From this circuit we see that such a group must contain $CX_{12}, CX_{23}, CCX_{124}, CCX_{234}$, as well as X_1, X_2, X_3, X_4 . By enforcing the closure property of a group (Appendix A), we also see that it contains the four X Paulis, the Cliffords CX_{ij} for $i < j$, and the non-Clifford gates CCX_{ij4} where $i < j$. Denoting each of these as individual groups, we have;

$$\begin{cases} H = \langle X_1, X_2, X_3, X_4 \rangle \\ K_1 = \langle CX_{12}, CX_{13}, CX_{14}, CX_{23}, CX_{24}, CX_{34} \rangle \\ K_2 = \langle CCX_{124}, CCX_{134}, CCX_{234} \rangle \end{cases} \quad (1)$$

First, we see that $H \cong \mathbb{Z}_2^4 \implies |H| = 16$ since every element commutes and is order two. K_1 is non-Abelian, with each generator representing a lower triangular 4×4 matrix with a single 1 below the diagonal, over the field $\mathbb{F}_2$. Since there are six entries that are either 0 or 1, this gives us a group of order $|K_1| = 2^6 = 64$. Similarly, $K_2 \cong \mathbb{Z}_2^3$, so $|K_2| = 8$, since the three CCX generators share the same target, commute with one another, and each has order two. Note however, $H \rtimes K_1$ is not normal in the full group: conjugation by K_2 generators can introduce K_2 factors. The full group then takes the form (Appendix A)

$$G_{FA} = (H \rtimes K_1) \bowtie K_2 \quad (2)$$

with $|G_{FA}| = 16 \cdot 64 \cdot 8 = 2^{13} = 8192$, and where $\bowtie$ is the knit product, which contains both K_2 and $H \rtimes K_1$ as subgroups, the only element in common is the identity, and every element in G_{FA} can be decomposed as the product of an element in $H \rtimes K_1$ with an element in K_2 [1].

Now we can consider a general element of the group, which will be represented by:

$$X_1^\alpha X_2^\beta X_3^\gamma X_4^\delta (CX_{12})^a (CX_{13})^b (CX_{14})^c (CX_{23})^d (CX_{24})^e (CX_{34})^f (CCX_{124})^p (CCX_{134})^q (CCX_{234})^r \quad (3)$$

where each power is either $\{0, 1\}$, giving 13 binary parameters. From this we can then treat the logical qubit as:

$$|\alpha, \beta, \gamma, \delta, a, b, c, d, e, f, p, q, r\rangle \quad (4)$$

The first four qubits carry the data, while the remaining nine ancilla qubits are initialized to $|+\rangle_{\text{anc}} = \frac{1}{\sqrt{2^9}} \sum_{\boldsymbol{\eta} \in \mathbb{F}_2^9} |\boldsymbol{\eta}\rangle$. Left multiplication by a gate L_q acts on the full normal-form coordinates as $(\mathbf{x}, \boldsymbol{\eta}) \mapsto (q(\mathbf{x}), \pi_{q, \mathbf{x}}(\boldsymbol{\eta}))$, where $\mathbf{x} \in \mathbb{F}_2^4$ are the data coordinates and $\boldsymbol{\eta} \in \mathbb{F}_2^9$ are the auxiliary coordinates. Since the auxiliary sector is prepared in $|+\rangle$, it is invariant under the conditional permutations $\pi_{q, \mathbf{x}}$, and the logical action reduces to the data-coordinate transformation $\mathbf{x} \mapsto q(\mathbf{x})$ [1]. Throughout this section, products of logical operators are written in standard right-to-left application order. By applying the logical operators for each generator, we arrive at the conclusion that:

$$L_{FA} = L_{CX_{12}} L_{CX_{23}} L_{CCX_{234}} L_{CX_{12}} L_{CCX_{124}} \quad (5)$$

implements the logical operation equivalent to the full adder, when our initial qubit is initialized to $|\alpha, \beta, \gamma, 0\rangle \otimes |+\rangle_{\text{anc}}$. We verify this rigorously in Appendix A.

B. Modular GSC Implementation of the Full Adder

We begin by considering five $\mathbb{Z}_2$ surface code patches P_1, P_2, P_3, P_4, P_a associated with the four logical qubits

initialized to $|a\rangle, |b\rangle, |c_{in}\rangle, |0\rangle$, and one ancilla qubit which will be used to implement the CX gates via lattice surgery. To start, we make use of the GSC $G_{CCX} = (H \rtimes K_1) \rtimes K_2$ which has order $|G_{CCX}| = 64$ as analyzed in [1] two times, one for the CCX_{124} gate and one for the CCX_{234} gate. For the CX gates, we will use the method of lattice surgery as described in [6], both of which we develop generally before implementing for the full adder.

1. Implementing the CCX gate

We closely follow the extension and splitting used in [1] to develop the following process:

Outline: Let P_i, P_j, P_k be patches of $\mathbb{Z}_2$ surface code, and denote logical states by $|x\rangle_i, |y\rangle_j, |z\rangle_k$ supported on these patches. We can then implement the operation:

$$U_{CCX} : |x\rangle_i \otimes |y\rangle_j \otimes |z\rangle_k \mapsto |x\rangle_i \otimes |y\rangle_j \otimes |z \oplus xy\rangle_k \quad (6)$$

by using the logical operation L_{CCX} in the G_{CCX} code, along with the sequence of splittings and extensions:

$$\text{Ext}_{H, K_1} \rightarrow \text{Ext}_{H \rtimes K_1, K_2} \rightarrow L_{CCX} \rightarrow \text{Spl}_{H \rtimes K_1, K_2} \rightarrow \text{Spl}_{H, K_1} \quad (7)$$

Where the three qubits for sectors K_1 and K_2 are initialized to $|+\rangle$, so that when applying L_{CCX} they do not become entangled, and are then removed when carrying out the splitting operation, turning the three patches back into $\mathbb{Z}_2$ surface code patches.

2. Implementing the CX gate

Following lattice-surgery CNOT protocols [6], a logical CX_{ij} from control patch P_i to target patch P_j can be implemented using an ancilla patch P_a . The ancilla is initialized in the logical state $|+\rangle_a$. The protocol measures two joint logical parities by merge/split lattice surgery and then measures the ancilla patch.

Let the measurement outcomes be represented as bits, with 0 corresponding to eigenvalue +1 and 1 corresponding to eigenvalue -1.

Step 1: Prepare the ancilla patch in $|+\rangle_a$.

Step 2: Measure the joint logical parity $Z_L^{(i)} Z_L^{(a)}$ by a rough-boundary merge of P_i and P_a , followed by d rounds of syndrome extraction and then a split. Denote the resulting parity bit by $m_Z \in \{0, 1\}$.

Step 3: Measure the joint logical parity $X_L^{(a)} X_L^{(j)}$ by a smooth-boundary merge of P_a and P_j , followed by d rounds of syndrome extraction and then a split. Denote the resulting parity bit by $m_X \in \{0, 1\}$.

Step 4: Measure the ancilla patch P_a in the logical Z basis. Denote this measurement outcome by $m_a \in \{0, 1\}$.

Step 5: Apply, or track in the Pauli frame, the byproduct corrections

$$\left(Z_L^{(i)}\right)^{m_X} \left(X_L^{(j)}\right)^{m_Z \oplus m_a}. \quad (8)$$

After these corrections, the net logical operation on P_i and P_j is

$$CX_{ij} : |x\rangle_i |y\rangle_j \mapsto |x\rangle_i |x \oplus y\rangle_j. \quad (9)$$

3. Implementing the Full Adder

Now that we have developed the methods for implementing the CCX and CX gates, we must ensure the geometry of the patches results in the required boundaries for merging and extension, with the proposed set up shown in Fig. 2. For the CX gates, we must have the boundaries of patch 1, 2, and 3 sharing a boundary with the ancilla. We must also have patches 1, 2, 3 and 4 sharing a boundary for the G_{CCX} extensions. Then using the described processes, we implement the Full adder by the sequence:

$$U_{FA} = U_{CX_{12}} U_{CX_{23}} U_{CCX_{234}} U_{CX_{12}} U_{CCX_{124}} \quad (10)$$

Where $U_{CCX_{ij4}}$ represents the CCX gate process, and CX_{ij} represents the CX gate process, resulting in a full quantum adder.

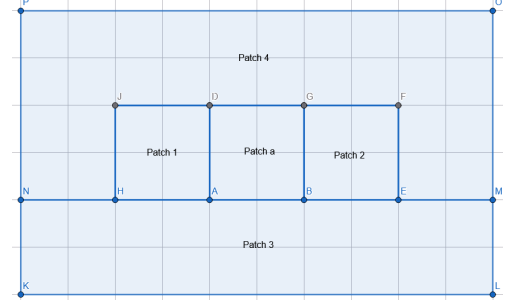


FIG. 2. Schematic compact layout for the modular full-adder construction. The subregions indicate the required adjacencies for lattice surgery and extension/splitting operations, not an exact metric edge count. In the resource estimates of Sec. IV, each active logical block is counted as a compact distance- d square patch with $2d(d-1)$ edge registers.

IV. QUANTUM ADDER CONSTRUCTION COMPARISONS

We now compare these two constructions in seven ways: qubit overhead, time depth, space-time volume, decoder complexity, error structure, scaling of qubits per edge, and overall implications for the decoder. The layout in Fig. 2 is used only as an adjacency schematic. The numerical prefactors below use a compact-square

benchmark: whenever an active logical block is treated as a distance- d patch, we count it as having $2d(d-1)$ edge registers. These leading-order estimates do not include elementary-gate depth for group multiplication within each edge register, syndrome-extraction scheduling overhead, routing, or hardware-specific implementation costs. Exact polygonal embeddings can change layout-dependent constants, but not the asymptotic scaling considered here. For a GSC based on a group G , each edge carries a $|G|$ -dimensional Hilbert space. A qubit representation therefore requires $q = \lceil \log_2 |G| \rceil$ physical qubits per edge; for the 2-groups considered below, this reduces to $q = \log_2 |G|$.

A. Qubit Overhead

Construction A - The G_{FA} code operates on a single GSC patch, and has order $|G_{FA}| = 2^{13}$, so each edge requires 13 physical qubits. The total is thus (including the leading order approximation):

$$Q_A = 13 \times 2d(d-1) = 26d^2 - 26d \approx 26d^2 \quad (11)$$

This count is static for the entire circuit, as this GSC allows the full adder to be implemented through a sequence of transversal left-multiplication operators, L_g .

Construction B - This modular implementation of the full adder changes throughout the procedure, and so the number of physical qubits varies depending on the operation. We begin with five $\mathbb{Z}_2$ surface code patches, four for the data qubits and one for the ancilla qubit, each requiring one qubit per edge. We now identify the number of physical qubits for each phase:

1. *Idle* - All five patches are $\mathbb{Z}_2$ surface codes, so the number of qubits is:

$$Q_{idle} = 5 \times 2d(d-1) = 10d^2 - 10d \approx 10d^2 \quad (12)$$

2. *During CCX* - When applying the U_{CCX} operation, the three participating $\mathbb{Z}_2$ logical degrees of freedom are transferred, through extension, into a single compact distance- d G_{CCX} patch with order $|G_{CCX}| = 2^6$, requiring 6 qubits per edge. Under the compact-square benchmark described above, this active G_{CCX} patch has $2d(d-1)$ edge registers. The remaining two patches stay as $\mathbb{Z}_2$, so the total qubit count is:

$$Q_{CCX} = (6+2) \times 2d(d-1) = 16d^2 - 16d \approx 16d^2. \quad (13)$$

3. *During CX* - Since this process does not change the $\mathbb{Z}_2$ patches, the total number of qubits will be the same as the idle case, $Q_{CX} = 10d^2 - 10d \approx 10d^2$

Under this leading-order layout assumption, the maximum qubit count for Construction *B* is smaller than that of Construction *A*, giving a spatial reduction factor of approximately $26/16 = 1.625$.

B. Time Depth

Each construction has a time depth measured in units of d syndrome-extraction rounds [6], which is the minimum number of repeated measurements needed for reliable error correction at distance d .

Construction A - This construction consists of five transversal left-multiplication gates, (5). Each of these gates has a circuit depth of one, and between each gate there are d rounds of syndrome extraction for fault-tolerant error correction, so the total time depth is:

$$T_A \approx 5d \quad (14)$$

Construction B - This time cost includes the U_{CCX} operation (extension and splitting) as well as the U_{CX} operation (lattice surgery). For each CCX gate, we have the following five steps:

1. The two extensions, Ext_{H,K_1} and $\text{Ext}_{H \rtimes K_1, K_2}$, require the patch to be grown column by column, and so for a distance d patch there will be d measurement rounds.
2. The transversal L_{CCX} operation requires one gate step followed by d rounds of syndrome extraction.
3. The two splittings are the reverse of extensions, and so require d measurements to shrink the patch and measure out the ancilla qubits.

Thus the time cost per CCX operation is $5d$, resulting in a total of $10d$ from both. For each of the CX gates, the procedure takes approximately $3d$ rounds from the three operations, ZZ merge, XX merge, and the Z -basis measurement, with corrections. These three CX gates then contribute a time depth of $9d$ rounds, and so the total required is:

$$T_B \approx 19d \quad (15)$$

Comparing to construction *A*, construction *B* has a slowdown by a factor of $19/5 = 3.8$. However, if the CCX and CX operations could be parallelized, such as a CX gate being applied at the same time as an extension or splitting if they do not interfere geometrically or logically, then the time depth could possibly be reduced.

C. Space-Time Volume

Another important measure of comparison between codes is the space-time volume $V = Q \times T$ which scales as d^3 and is discussed extensively in [8]. To leading order, construction *A* has a space-time volume of:

$$V_A \approx 26d^2 \times 5d = 130d^3 \quad (16)$$

For Construction *B*, using the same leading-order layout assumption for the CCX phases, we have time depth

$10d$ and $16d^2$ physical qubits for the CCX gates, and time depth $9d$ and $10d^2$ physical qubits for the CX gates, giving:

$$V_B \approx 10d \times 16d^2 + 9d \times 10d^2 = 250d^3 \quad (17)$$

Under this convention, the leading-order space-time volume of Construction B is approximately $250/130 \approx 1.92$ times that of Construction A . Construction A has smaller leading-order space-time volume despite its group being larger by a factor of 128, because its fully transversal gates avoid the time overhead of extension, splitting, and lattice surgery required by Construction B . These estimates are leading-order qubit-round counts under the compact-square benchmark and do not include elementary-gate depth for group multiplication within each edge register, syndrome-extraction scheduling overhead, routing, or hardware-specific implementation costs. The complexity of the decoder, error threshold, and per-round syndrome-extraction circuit depth may dominate the practical feasibility. Thus we next analyze the decoder complexity.

D. Decoder Complexity

We now analyze the decoder complexity for both constructions, which can give insight into their practical feasibility.

Construction A - The G_{FA} GSC requires a decoder for the quantum double model of it, an order 2^{13} non-Abelian group, which must simultaneously handle two types of excitations:

1. *Flux errors*, labeled by conjugacy classes of G_{FA} . These can be moved deterministically using the spacetime ribbon operators described in Section IV A of [1]. Note that the number of distinct flux types equals the number of conjugacy classes of the group.
2. *Charge errors*, labeled by irreducible representations of G_{FA} . Since it is non-Abelian, some of the irreps are multi-dimensional, meaning charges carry internal states that are rearranged by vertex measurements [1].

A favorable property of G_{FA} is that it is a p -group, a group of order p^a where p is prime (2 in this case), and thus it will be nilpotent [9]. From [1], it is noted that for nilpotent groups, non-Abelian charge movement succeeds in a bounded number of attempts, avoiding the unbounded probabilistic overhead that can arise for general non-Abelian groups. Nevertheless, the number of distinct anyon types in the quantum double of G_{FA} scales with both the number of conjugacy classes and the structure of their centralizers [10]. Due to the large size of our group, the conjugacy class structure and centralizers will be complex, and so far no efficient decoding algorithm exists for general non-Abelian

groups. Furthermore due to its size, the qubit-level implementation overhead of syndrome extraction scales with $q = \log_2 |G_{FA}| = 13$, since each edge carries a $|G_{FA}|$ -dimensional system, introducing additional physical error opportunities.

Construction B - The modular construction benefits from separating the quantum adder into simpler phases, which would involve a simpler decoder.

1. *Between Operations*: Standard $\mathbb{Z}_2$ surface code decoding, which can be done with near-optimal decoders achieving thresholds on the order of 1% [2].
2. *During lattice surgery*: The merge and split operation introduce correlated errors at the boundary between patches, but the underlying code remains in $\mathbb{Z}_2$ and so efficient decoders for lattice surgery can be used [8], [2], [6].
3. *During G_{CCX} extension*: We require a decoder for the GSC G_{CCX} , which has 2^6 elements. Although it is still non-Abelian, it is 2^7 times smaller than G_{FA} and so the syndrome-extraction circuits involved controlled operations on 6 qubit systems instead of 13. Note that it is also nilpotent since it is a p -group, so charge movement is achievable in a fixed amount of attempts.
4. *At phase boundaries*: When switching between the $\mathbb{Z}_2$ and G_{CCX} decoders, correlated errors from the extension and splitting must be handled. From [1], the charges appearing during extensions are constrained; they belong to irreps appearing in the induction of the trivial H -representation, and all such charges have a vacuum condensation channel at the $H - G$ interface. Thus they can always be removed by transporting them into the interface, though the transport operators have linear depth.

In summary, Construction A requires a single, complex decoder operating continuously on a large non-Abelian code, while Construction B requires coordination between well-understood $\mathbb{Z}_2$ decoders and a simpler G_{CCX} decoder. While neither have an established error threshold at present, the smaller group order and the modular decoder architecture make Construction B substantially more tractable for a theoretical analysis and practical implementation.

E. Error Structure and Transversality

We now analyze how errors propagate in the two constructions, with particular attention to the role of full transversality in Construction A .

1. Construction A: Transversal gate phase

All five gates composing the full adder— $L_{CCX_{124}}$, $L_{CX_{12}}$, $L_{CCX_{234}}$, $L_{CX_{23}}$, and $L_{CX_{12}}$ —are transversal in Construction A. Each L_g acts as a tensor product $\bigotimes_e U_g^{(e)}$ of single-site unitaries on the $|G_{FA}|$ -dimensional edge register [1]. Transversality guarantees that a fault during the logical gate layer cannot propagate from one edge register to another: a single physical error within the implementation of $U_g^{(e)}$ may create a high-weight error within the 13-qubit register of edge e , but it cannot spread to a different edge register e' during the transversal layer. This is the strongest form of inter-edge fault isolation available during gate application.

In surface-code implementations, correlated data errors can arise from entangling operations used in syndrome extraction, lattice surgery, and other multi-qubit gate layers. In Construction A, this inter-edge noise channel is suppressed during the five transversal logical gate steps. During these steps, all multi-edge error correlations therefore originate exclusively from the syndrome extraction circuits.

2. Construction A: Syndrome extraction phase

The fault tolerance advantage of transversal gates does not extend to syndrome extraction. The plaquette stabilizer for a face f with v boundary edges is measured by a sequential controlled-multiplication circuit involving v controlled gates between an ancilla register and the data edges [1]. Each controlled gate acts on the full $|G_{FA}|$ -dimensional edge register, which is itself encoded in $q = 13$ physical qubits. A single fault within the qubit-level decomposition of such a gate can propagate to multiple edges in the extraction sequence, as analyzed in Proposition V.1 for the G_{CCX} case.

For the G_{FA} code, each controlled-multiplication gate decomposes into $O(\text{poly}(q))$ elementary two-qubit gates, increasing the number of fault locations per extraction round relative to both the $\mathbb{Z}_2$ surface code ($q = 1$) and the G_{CCX} code ($q = 6$). The fault propagation structure, however, is analogous: for the same sequential extraction schedule used in Proposition V.1, the stabilizer reduction argument gives an estimated effective error weight of $w_{\max} = \lfloor v/2 \rfloor$ after applying the plaquette stabilizer. On the square lattice ($v = 4$), this gives the same edge-register support estimate $w_{\max} = 2$. Thus the same distance scale, $d \geq 5$, would be sufficient to separate such local edge-register faults from logical operators. This support estimate does not by itself constitute a decoder for G_{FA} ; a full decoder analogous to the one developed for Construction B in Sec. V remains an open problem.

The error propagation profile of Construction A is therefore *bifurcated*: the five gate steps produce at most single-edge errors (no inter-edge propagation), while the d syndrome extraction rounds per gate step are the dom-

inant source of correlated multi-edge errors—the same bottleneck that limits the $\mathbb{Z}_2$ surface code.

3. Construction B: Extension, splitting, and lattice surgery

Construction B introduces two additional sources of correlated errors beyond the G_{CCX} bulk extraction circuits.

During the extension and splitting protocol, the code transitions through intermediate phases $\mathbb{Z}_2 \rightarrow H \rtimes K_1 \rightarrow G_{CCX}$, with each transition implemented by a sequence of controlled-measurement circuits at the interface between the two code domains [1]. A single fault at the interface can create correlated defects straddling the boundary, as analyzed in Sec. V E. While these are correctable for $d \geq 5$, the temporal overhead is significant: each extension or splitting requires d measurement rounds, and there are four such transitions per CCX gate (two extensions, two splittings).

During the lattice surgery CX gates, merge and split operations introduce correlated errors at the boundary seam between patches. These are handled by the standard $\mathbb{Z}_2$ lattice surgery decoder, but each merge/split operation adds d rounds of potentially correlated measurements.

4. Comparative assessment

At the level of syndrome-round counting, Construction A has five gate layers and approximately $5d$ extraction rounds, whereas Construction B has two G_{CCX} transversal gate layers and approximately $19d$ measurement rounds across all phases (extension, splitting, lattice surgery, and bulk extraction). Construction A therefore benefits from a $3.8\times$ reduction in temporal depth, with the additional structural advantage that none of its gate steps introduce inter-edge correlated errors.

This comparison favors Construction A in temporal depth and inter-edge fault isolation, but not necessarily in circuit-level fault rate, because each G_{FA} extraction round involves larger controlled group-multiplication circuits with more elementary fault locations per round than the corresponding G_{CCX} or $\mathbb{Z}_2$ circuits in Construction B. Whether the reduced number of rounds outweighs the increased per-round fault density depends on the physical error rate and the specific gate decomposition, and cannot be resolved without a full circuit-level threshold simulation.

We note that Manjunath et al. observe that for nilpotent groups—which include all 2-groups such as G_{FA} and G_{CCX} —charges can be moved in a fixed number of attempts [1], and recent work on the quantum double of D_4 has demonstrated non-Abelian charge-noise thresholds of $p_c \approx 0.159$ under standard MWPM, $p_c \approx 0.208$ under intrinsically heralded MWPM, and $p_c \approx 0.218$ under an optimal decoder, all with perfect syndrome measurements

[11]. These results suggest that GSC thresholds for small non-Abelian 2-groups can approach surface-code levels, but the extension to circuit-level noise with imperfect syndromes remains an important open problem.

F. Scaling of Physical Qubits per Edge

The comparisons in the preceding subsections treat the full adder as a fixed-size circuit. We now ask how the GSC resource cost scales as the target circuit grows, exposing a fundamental trade-off between gate simplicity and encoding overhead.

1. Edge dimension and group order

In a GSC built on a group G , each edge of the lattice hosts the group algebra $\mathbb{C}[G]$, a $|G|$ -dimensional Hilbert space [1]. Representing this space on qubits requires either a native $|G|$ -level qudit or $q = \lceil \log_2 |G| \rceil$ physical qubits per edge. In the 2-group examples studied throughout this paper, $|G|$ is always a power of two, so $q = \log_2 |G|$ exactly.

The edge dimension is fixed by the choice of group and is independent of the code distance d ; increasing d multiplies the number of edges but does not change the local Hilbert space. The total physical qubit count therefore factorizes as

$$Q = \underbrace{q}_{\text{edge dim.}} \times \underbrace{2d(d-1)}_{\text{edges per patch}}, \quad (18)$$

so the edge dimension acts as a multiplicative overhead that amplifies every distance-dependent cost.

2. Exponential growth of the edge dimension

For the family of multi-controlled- X groups introduced in [1], the group $G_{C^n X}$ is generated by the Pauli operators $X_1, \dots, X_n$ together with the n -controlled- X gate $C^n X_{(1\dots n), n+1}$ acting on $n+1$ qubits. Although X_{n+1} is not listed as an independent generator, it is contained in the generated group. Conjugating the control Paulis by $C^n X$ produces controlled- X operations on the target at successively lower levels, and iterating these commutators eventually yields the target Pauli X_{n+1} . The closure introduces all controlled- X gates with k controls for $1 \leq k \leq n$, with $\binom{n}{k}$ gates at each level k , giving a nested semidirect product of order [1]

$$|G_{C^n X}| = 2^{(n+1) + \sum_{i=1}^n \binom{n}{i}} = 2^{2^n + n}. \quad (19)$$

The number of physical qubits per edge is therefore

$$q = \log_2 |G_{C^n X}| = 2^n + n, \quad (20)$$

which grows *exponentially* in the number of controls n . The group order $|G_{C^n X}| = 2^{2^n + n}$ itself grows doubly exponentially, but since each edge requires only $\log_2 |G|$ qubits, the physical qubit cost per edge is governed by the singly exponential factor $2^n + n$.

Table I lists the edge dimensions for the $G_{C^n X}$ family and for the groups used in this paper.

TABLE I. Edge dimension $q = \log_2 |G|$ for selected GSCs. The $G_{C^n X}$ rows refer to the left-multiplication construction of multi-controlled- X gates [1]. Automorphism gates can give additional transversal logical operations beyond those listed in the “Role” column.

Code	n	$ G $	q	Role
$D_4 \cong G_{CX}$	1	8	3	left-action CX
G_{CCX}	2	64	6	left-action CCX
$G_{C^3 X}$	3	2^{11}	11	left-action $C^3 X$
$G_{C^4 X}$	4	2^{20}	20	left-action $C^4 X$
$\mathbb{Z}_2$	—	2	1	standard surface code
$\mathbb{Z}_2^2$	—	4	2	two-qubit abelian GSC
G_{FA} (this paper)	—	2^{13}	13	monolithic full-adder group

The growth from $n = 2$ to $n = 4$ is stark: the edge dimension increases from 6 to 20, and by $n = 5$ each edge would require 37 physical qubits. This exponential scaling means that encoding an entire multi-qubit reversible circuit as a single monolithic GSC quickly becomes infeasible. A larger circuit such as a 4-bit ripple-carry adder, which chains four full adders in sequence, would generally be expected to produce a larger closure group than the single-full-adder group G_{FA} , but determining the exact edge dimension requires a separate group-enumeration computation. The practical strategy is to chain multiple modular G_{CCX} patches sequentially via extension and splitting, keeping the per-patch edge dimension at $q = 6$ and paying the code-switching overhead once per non-Clifford gate.

3. The group algebra as a unary encoding

The scaling in Eq. (20) admits a representation-theoretic interpretation. The GSC Hilbert space on each edge is the regular representation $\mathbb{C}[G]$, in which every group element g labels a distinct basis vector $|g\rangle$. On a single edge, left multiplication by any $h \in G$ acts as the permutation $|g\rangle \mapsto |hg\rangle$. Logical left multiplication is implemented as a product of such single-edge permutations over the appropriate boundary support, so it is transversal over that support. Complex multi-qubit reversible operations—including CCX gates and arbitrary compositions thereof—reduce to group shifts in this basis.

This is analogous to a unary encoding of classical data: an integer $m < N$ is stored in the position of a single token among N slots, rather than as a $\lceil \log_2 N \rceil$ -bit binary string. Arithmetic on unary data is trivial at the level

of state permutations, but this simplicity is achieved by enlarging the state space. The analogy for GSCs is not a literal unary qubit encoding; rather, the regular representation has one basis state for each group element. A GSC converts a non-Clifford gate from a nontrivial multi-qubit operation into a group shift on this enlarged local Hilbert space, represented on qubits using $q = \log_2 |G|$ physical qubits per edge in the 2-group cases considered here.

The encoding and decoding overhead—the extension and splitting protocols that transfer information between $\mathbb{Z}_2$ surface codes and the G -GSC—is the cost of translating between the binary and unary representations. This cost is paid once per entry into the non-Clifford phase and is amortized across all transversal gates applied within that phase. Construction B exploits this by performing the extension/splitting overhead twice (once per CCX gate) while executing three CX gates in the cheaper $\mathbb{Z}_2$ lattice surgery framework.

4. Comparison with magic state distillation scaling

The overhead of magic state distillation scales qualitatively differently. The number of raw magic states required to distill one T -state to accuracy ϵ is $O(\log^\gamma(1/\epsilon))$ with $\gamma = \log_2 3 \approx 1.585$ [12], and recent protocols achieve asymptotically constant space overhead [13]. For fixed low-arity gates such as CCX , a single Toffoli can be implemented by consuming $O(1)$ T -states or a single $|CCZ\rangle$ resource state via gate teleportation [14, 15]; the number of high-fidelity resource states consumed per logical gate is $O(1)$. However, the distillation overhead required to prepare each such resource state still depends on the target accuracy ϵ .

For higher-arity gates such as C^nX with large n , however, the comparison is more nuanced. Standard decompositions of C^nX into Toffoli gates require $O(n)$ Toffoli gates with ancillas [16], so the magic-state cost grows linearly in n . The GSC edge dimension, by contrast, grows exponentially as $2^n + n$. Both overheads therefore increase with gate arity, but with qualitatively different scaling.

The two cost structures cross: for low-arity gates applied many times in sequence on the same logical qubits, the GSC overhead can be competitive because the extension/splitting cost is amortized; for high-arity gates or circuits requiring only a few non-Clifford operations, magic state distillation is more efficient. For the full adder studied here, the relevant comparison is between the $6\times$ qubit overhead per edge of the G_{CCX} code and the space-time volume of a $|CCZ\rangle$ factory [15]. A detailed numerical comparison at specific physical error rates and code distances is left for future work, but the structural trade-off is clear: the GSC approach is favored in regimes where many sequential non-Clifford gates act on the same logical qubits, while magic state distillation is favored for shallow circuits with sparse non-Clifford

content.

TABLE II. Leading-order quantitative comparison of the monolithic (A) and modular (B) full adder constructions at code distance d . The Construction B qubit counts assume that the active G_{CCX} region is counted as a single distance- d patch during each CCX operation.

Metric	A	B
Group order	$2^{13} = 8192$	$2^6 = 64$
Qubits per edge	13	1–6
Peak qubits	$26 d^2$	$16 d^2$
Idle qubits	$26 d^2$	$10 d^2$
Time depth	$5d$	$19d$
Space-time volume	$130 d^3$	$\sim 250 d^3$
Gate transversality	All five gates	CCX only
Decoder type	non-Abelian	abelian + non-Abelian
Nilpotent group	Yes	Yes
Edge-register width	13 qubits	1–6 qubits

G. Implications for the Decoder

Table II summarizes the fixed-size comparison. Under the leading-order compact-square convention, Construction A has lower space-time volume, while Construction B has a smaller non-Abelian group, lower edge-register width during non-Abelian phases, and a modular decoder architecture. The remainder of the paper therefore focuses on Construction B, where a concrete single-fault decoder can be analyzed.

V. DECODER FOR CONSTRUCTION B

We now construct a single-fault decoder for Construction B of the full quantum adder, by using standard decoders for the $\mathbb{Z}_2$ phases of the circuit, the extending and splitting transitions, as well as constructing a decoder for the G_{CCX} implementation of the CCX gate. The input for this decoder will be syndromes from plaquette measurements, which give holonomy outcomes (flux), and vertex measurements, which give irreps (charge) [1]. We prove that in the single-fault regime, the decoder corrects all errors without producing logical failures.

A. Fault Propagation

We work on a square lattice with the number of edges connecting to one vertex being $v = 4$, and each edge hosts a $|G_{CCX}| = 2^6 = 64$ dimensional register encoded by $q = 6$ physical qubits [1]. The syndrome-extraction circuits involve controlled multiplication gates whose qubit-level decomposition introduces multiple fault locations per gate, with the following proposition establishing a key bound:

Proposition V.1 *After stabilizer reduction, a single circuit-level fault during either a plaquette or vertex syndrome-extraction circuit produces an error of edge-register weight at most $w_{\max} = 2$.*

Proof. Plaquette Faults. The plaquette stabilizer for face f with boundary edges $e_1, \dots, e_v$ is measured by initializing an ancilla register to the identity element $|e\rangle$ and applying controlled-multiplication gates $C_{e_i \rightarrow \text{ancilla}}$ for $i = 1, \dots, v$ in sequence, where each data edge controls the ancilla target [1].

A Z -type fault on the ancilla propagating through the remaining gates at position j corrupts data edges $e_j, \dots, e_v$, producing $v - j + 1$ edge register errors. Applying the plaquette stabilizer B_f transforms this into an equivalent error of weight $v - (v - j + 1) = j - 1$. The weight is therefore $\min(v - j + 1, j - 1)$ for $j = 1, \dots, v$. For $v = 4$, we have that $\min(5 - j, j - 1)$ is maximized at $j = 3$, giving $w_{\max} = 2$. An X -type fault on the ancilla does not propagate to data edges, since the ancilla is the target, and faults within the qubit-level decomposition of a single $C_{e_i \rightarrow \text{ancilla}}$ can corrupt at most the ancilla register and the single data register e_j . Since the remaining controlled-multiplication gates act on different data registers, then it cannot corrupt other data edges.

Vertex Faults. The vertex stabilizer uses controlled-multiplication gates $C_{\text{ancilla} \rightarrow e_i}$, where the ancilla is the control [1]. By the dual argument, X -type faults on the ancilla propagate while Z -type faults do not, and thus stabilizer reduction again yields $w_{\max} = 2$.

Since correcting up to $w_{\max} = 2$ edge register errors requires distance to satisfy $\lfloor \frac{d-1}{2} \rfloor \geq w_{\max}$, taking $w_{\max} = 2$ then yields $d \geq 5$.

B. Algebraic Properties of G_{CCX}

The decoder relies on several algebraic properties of G_{CCX} , which we state as propositions and verify computationally.

Proposition V.2 (Self-inverse Conjugacy Classes) *G_{CCX} has 16 conjugacy classes, all of which are self-inverse: for every class C and every $g \in C$, then $g^{-1} \in C$.*

Proposition V.3 (Self-dual irreps) *All 16 irreducible representations of G_{CCX} are self-dual: $R \cong R^*$ for every irrep R .*

Proposition V.4 (Unique Vacuum Fusion Channel) *For every irrep R of G_{CCX} , the tensor product $R \otimes R$ contains the trivial representation 1 with multiplicity exactly 1.*

All of these propositions are justified in Appendix B.

C. Flux Decoder

1. Plaquette Measurement, Errors, and Conjugacy Classes

For a plaquette whose edge qubits are g_1, g_2, g_3 , and g_4 (traversed with consistent orientation), the holonomy measurement will be $g_1^{-1}g_2g_3g_4^{-1}$ [1]. In the error free case this product is trivial, but in the case of an error on edge e , we get the group element $m_f \neq 1$ for face f . The two faces bordering e receive measurements related by inversion (m_f and m_f^{-1}), and gauge transformations conjugate the measurement without changing its conjugacy class. By Proposition V.2, the class label is the same on both sides of e , enabling pairwise matching.

2. Flux Decoder Algorithm

The self-inverse property enables minimum weight pairwise matching (MWPM) [17] within each conjugacy class:

1. Collect all plaquette defects and label each by its conjugacy class.
2. Group defects with the same class label.
3. Within each group, perform MWPM with virtual boundary vertices (weight equal to the distance to the nearest boundary) [17]
4. Correct each matched pair by applying a ribbon operator [1, 18] along the matching path.

We now bound the number of flux defects created by a single fault. By Proposition V.1, the stabilizer induced error lies on at most $w_{\max} = 2$ edge registers. In the plaquette extraction circuit, these are consecutive boundary edges of the plaquette f at which the fault occurred, since the reduction maps the error to consecutive positions in the extraction sequence. Two consecutive boundary edges of a plaquette on a square lattice share exactly one face, f , and each borders one additional face, for a total of at most 3 distinct plaquette defects. Because these defects arise from at most two consecutive edge-register errors in a single plaquette-extraction circuit, their spatial support lies in a contractible neighborhood of diameter strictly less than d for $d \geq 5$. The corresponding correction ribbon therefore cannot implement a logical operator.

D. Charge Decoder

1. Vertex Measurement, Errors, and Irreducible Representations

The vertex stabilizer A_v^g acts by left-multiplying by g on outgoing edges and right-multiplying by g^{-1} on

incoming edges [1]. Without errors, the measurement yields the trivial representation, whereas an error on edge e creates excitations at both endpoints, labeled by irreps.

By Proposition V.3, all irreps of G_{CCX} are self-dual, so both endpoints of an error edge carry the same representation label R . This follows because two representations at the vertices of e are dual to each other, R and R^* , due to the opposite orientations: one vertex sees the error via left-multiplication, while the other via right-multiplication by the inverse. Self-duality collapses these to the same label, and the representation label is invariant under gauge transformations since characters are class functions [19].

2. Charge Decoder Algorithm

The self-duality and unique fusion (Proposition V.4) enable MWPM for charges:

1. Collect all vertex defects and label each by its irrep R .
2. Group vertices with the same irrep label.
3. Perform MWPM within each group, with virtual boundary vertices.
4. Transport one charge along the matching path using the charge-movement procedure of Ref. [1]. For the nilpotent groups considered here, this movement succeeds in a bounded number of attempts. The matched pair is then fused in the vacuum channel $\mathbf{1} \subset R \otimes R$.

For the single-edge faults considered in the single-fault regime, the two endpoint charges are created in the vacuum fusion channel. Together with the unique-vacuum-channel property of Proposition V.4, this makes the local single-fault correction unambiguous. By the same geometric argument as for flux, a single fault creates at most three vertex defects on consecutive edges sharing a common vertex, and for $d \geq 5$ no non-contractible loop is produced.

E. Extension and Splitting Interface Decoder

1. Intermediate Phases

During the extension/splitting protocol, the code passes through intermediate phases $\mathbb{Z}_2 \rightarrow H \rtimes K_1 \rightarrow G_{CCX} \rightarrow H \rtimes K_1 \rightarrow \mathbb{Z}_2$. We require decoders for each phase (which we have done for G_{CCX} , and is well established for $\mathbb{Z}_2$), and so proceed with $H \rtimes K_1$:

Proposition V.5 *The group $H \rtimes K_1$ has 17 conjugacy classes, all self-inverse, and 17 irreps, all self-dual. For every irrep R , the fusion $R \otimes R$ contains $\mathbf{1}$ with multiplicity 1.*

The justification is given in Appendix B.

These properties ensure that the classwise MWPM flux decoder and charge decoder apply to the $H \rtimes K_1$ phase in the same manner as to G_{CCX} .

2. Faults at the Interface Boundary

A single circuit-level fault during syndrome extraction near the interface produces at most $w_{\max} = 2$ edge register errors on consecutive edges. When one of the affected edges is a boundary edge e connecting vertex v_A in the lower order region (group A) to vertex v_B in the higher order region (group B), with $A \hookrightarrow B$, two scenarios arise depending on whether the second error edge e' lies in region A or region B .

In both scenarios, the error on the boundary edge e creates correlated defects at v_A and v_B : the vertex stabilizer at v_A detects the A component of the error, while v_B detects the full B component. The error on the bulk edge e' creates defects entirely within whichever region contains it, and is handled by that region's bulk decoder. The boundary edge e is corrected by the following interface protocol:

1. *Identify cross-interface defects:* Any defect pair (v_A, v_B) on adjacent vertices connected by a boundary edge, with v_A and v_B in different code domains, is flagged as cross-interface.
2. *Determine correction from the larger domain:* The B syndrome at v_B determines a correction operator C supported on the boundary edge register e . The vertex stabilizer of B restricted to the subgroup A coincides with the vertex stabilizer of A , which is guaranteed by the compatibility of the embedding $A \hookrightarrow B$ with the GSC stabilizer structure [1]. Therefore, any operator C that restores the B stabilizer at v_B automatically restores the A stabilizer at v_A upon restriction. This holds even when the error element on e lies in $B \setminus A$, since the correction is determined entirely from the B -side syndrome. Moreover, because C is supported on the single boundary edge e , it cannot create new defects in the A bulk.
3. *Apply correction:* The operator C , supported on the boundary edge e , resolves the cross-interface defects simultaneously.

The combined correction from the interface protocol and the bulk decoder involves at most 2 edge registers total. Together with the at most 3 flux or charge defects created by these two errors, the combined fault and correction support is contained in a local boundary neighborhood of weight at most $2w_{\max} = 4 < d$. Thus the support is contractible and cannot implement a logical operator.

The same argument applies to splitting, which is the time-reversal of extension: the splitting protocol measures out ancilla qubits and shrinks the patch, with the

same syndrome-extraction circuits run in reverse temporal order. A single fault during splitting produces the same type of localized errors on the boundary and adjacent bulk registers, correctable by the same protocol.

F. Single-Fault Correctness

We now combine these results to establish a single-fault correctness for Construction B:

Theorem V.6 *For code distance $d \geq 5$, any single fault occurring during syndrome extraction, extension, splitting, or logical gate application in Construction B produces a set of defects that can be corrected by the decoder described above, without resulting in a logical error.*

Proof. We verify this for each of the phases:

1. *Fault During $\mathbb{Z}_2$ bulk:* Between gate operations, all patches carry standard $\mathbb{Z}_2$ surface codes. A single fault is corrected by MWPM decoding [2, 17], and no logical error occurs for $d \geq 3$.
2. *Fault during lattice surgery CX:* The CX gates are implemented by standard $\mathbb{Z}_2$ surface-code lattice surgery [6] using local ZZ and XX merge measurements, followed by measurement of the ancilla patch. Under the standard syndrome-extraction schedule for the $\mathbb{Z}_2$ surface code, in which the CNOT ordering is chosen so that single-ancilla faults produce data errors on at most two adjacent edges [2, 17], a single circuit-level fault during these operations induces either a measurement error or a data error supported on at most two adjacent edges near the merge/split seam. The standard spacetime MWPM decoder for surface-code lattice surgery corrects such local defects. Since the induced data support is contained in a contractible neighborhood of diameter less than d for $d \geq 5$, the correction cannot produce a nontrivial logical operator.
3. *Fault during G_{CCX} :* By Proposition V.1, a single fault produces at most $w_{\max} = 2$ edge register errors, creating at most 3 flux or charge defects on consecutive faces or vertices. The classwise MWPM decoder constructed in Secs. V C–V D corrects these on at most 2 edge registers, so for $d \geq 5$ the correction is contractible and no logical error occurs.
4. *Fault during extension and splitting:* A fault within a bulk region ($\mathbb{Z}_2$, $H \rtimes K_1$, or G_{CCX}) is handled by the corresponding bulk decoder. A fault near an interface is handled by the subgroup compatible boundary rule of Sec. V E: the correction is chosen on the larger-group boundary register and therefore restores both the larger-group syndrome and its subgroup-restricted interface syndrome. Since

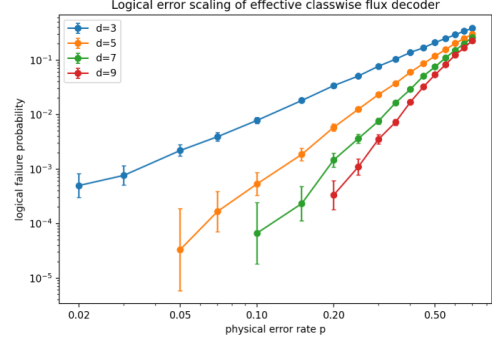


FIG. 3. Logical failure probability of the effective classwise MWPM flux decoder under the code-capacity noise model for distances $d = 3, 5, 7, 9$ with 30,000 trials per point.

the correction remains supported in the same local boundary neighborhood as the fault, it cannot create a noncontractible chain for $d \geq 5$.

VI. NUMERICAL SIMULATION

We test the classwise MWPM flux decoder of Secs. V C in the G_{CCX} bulk phase. The Monte Carlo simulations evaluate the decoder under the induced class-chain noise model, while the two-edge diagnostic compares this surrogate with the exact group-valued face-flux syndrome.

A. Code Capacity Model

Each supported edge independently suffers an error with probability p , where the error element is sampled uniformly from the 63 non-identity elements of G_{CCX} and assigned to its conjugacy class. Syndrome extraction is perfect, and the decoder runs independent MWPM per conjugacy class. A logical failure occurs when any class chain crosses the logical cut after correction.

Fig. 3 shows the logical failure probability for code distance $d \in \{3, 5, 7, 9\}$ with 30,000 trials per point. At every sampled error rate, increasing distance consistently suppresses the failure probability across the full range $p \in [.02, .70]$. The high apparent per-edge threshold is a consequence of class dilution: each conjugacy class sees an effective error rate $p_C = p \cdot \frac{|C|}{63}$, where $|C|$ is the class size. The largest class has 8 elements, giving $p_C = \frac{8}{63}p \approx .13p$. Since each class is decoded as an independent $\mathbb{Z}_2$ like matching problem with threshold 10.9% [17], the per-edge threshold of the classwise decoder is expected to be significantly higher than the $\mathbb{Z}_2$ value due to class dilution. This per edge advantage is offset by the $6\times$ qubit overhead per edge relative to $\mathbb{Z}_2$.

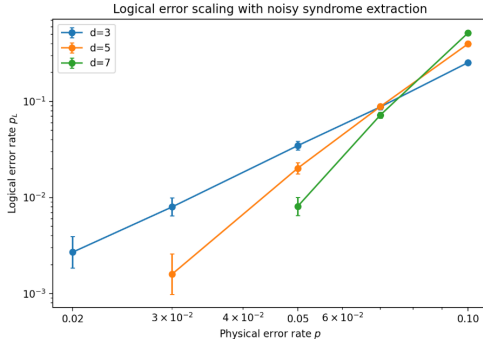


FIG. 4. Logical failure probability of the effective classwise MWPM flux decoder under the phenomenological class-chain noise model.

B. Phenomenological Monte Carlo

We extend to a phenomenological noise model: data errors occur at rate p per edge per round, syndrome measurements are independently flipped with probability $q = p$, and $r = d$ rounds of noisy extraction are followed by one noiseless anchor round. The decoder performs classwise MWPM in the resulting three-dimensional spacetime syndrome history, matching syndrome-change events with Manhattan-distance weights. Timelike only matched pairs produce no spatial correction.

Fig. 4 shows results for $d \in \{3, 5, 7\}$ with 10,000 trials per point. At low error rates, increasing distance suppresses the logical failure probability with $d = 7$ achieving the lowest failure rate. Near $p \approx .07$, the curves cross, indicating that the decoder has exceeded its effective pseudo-threshold under this model. The pseudo-threshold is a per-edge register quantity elevated by class dilution, and does not directly imply a per physical qubit advantage over $\mathbb{Z}_2$, since each G_{CCX} edge register carries 6 physical qubits. The $\mathbb{Z}_2$ phenomenological threshold of approximately 2.9% [20] should be read as a reference scale, not as evidence of a threshold advantage.

C. Two-Edge Diagnostic

The preceding simulations validate the decoder under the class-chain noise model; we now examine how closely this surrogate captures the exact group-valued syndrome. The classwise decoder assigns each edge error independently to a conjugacy class, but when two errors border the same face, their group-valued product can lie in a conjugacy class different from either individual error class, producing a syndrome the per-edge class assignment does not capture.

We quantify this exhaustively at $d = 5$. For every pair of supported edges sharing a face (120 pairs), we inject all $63^2 = 3,969$ ordered nontrivial element combinations, totaling 476,280 configurations, decoded under two syndrome methods: the class-chain syndrome (each edge in-

dependently assigned to its error's conjugacy class) and the exact group-valued face-flux syndrome (the oriented holonomy product projected to a conjugacy class). Under the class-chain syndrome, the decoder produces zero logical failures. Under the face-flux syndrome, 98.4% of configurations produce a syndrome disagreeing with the per-edge class assignment, and 28.1% result in logical failure.

These percentages are uniform-combinatorial diagnostic rates, not operational failure probabilities; a physical error channel would induce a nonuniform distribution over group elements whose failure rate depends on the microscopic implementation. The diagnostic quantifies the intrinsic vulnerability of the classwise approximation to inter-class coupling, and identifies a concrete target for future work: a group-valued MWPM decoder that matches on exact face-flux products rather than per-edge class labels.

VII. CONCLUSION AND DISCUSSION

We have presented a task-level case study of quantum arithmetic in the group surface code framework, constructing two implementations of a full quantum adder: a monolithic Construction A using a single G_{FA} code of order 2^{13} , and a modular Construction B using two G_{CCX} codes of order 2^6 combined with lattice surgery. Under a compact-square resource benchmark, Construction A has smaller leading-order space-time volume ($130d^3$ vs. $250d^3$) through fully transversal gate application, but requires a decoder for a group 128 times larger. Construction B trades this for a modular decoder architecture that combines well-understood $\mathbb{Z}_2$ decoders with a smaller non-Abelian decoder, making it substantially more tractable. We developed a single-fault decoder for Construction B, proved single-fault correctness for $d \geq 5$, and analyzed specific numerical simulations for multiple faults under an effective class-chain noise model. These resource comparisons do not include elementary-gate depth for group multiplication, syndrome-extraction scheduling overhead, or hardware-specific costs, and do not establish a fault-tolerance threshold.

A. Comparison with Magic State Distillation

The standard route to a fault-tolerant CCX gate uses the $\mathbb{Z}_2$ surface code supplemented by magic state distillation or cultivation. A single CCX can be decomposed into $O(1)$ T gates, each requiring a distilled magic state [14, 21]. Recent cultivation protocols [3] have substantially reduced factory overheads, but they still require additional space-time volume and introduce a nonzero failure probability per supplied non-Clifford resource state.

Construction B avoids magic state distillation entirely: the CCX gate is implemented transversally within the G_{CCX} code, at the cost of the extension and splitting

protocol. The relevant trade-off is between the distillation overhead, which requires additional qubits, factory space-time volume, distillation failure probability, and the GSC overhead, which requires $6\times$ qubit cost per edge, the $19d$ time depth of the modular protocol, and the need for a non-Abelian decoder.

At current magic state cultivation efficiencies, the distillation route is likely more practical for individual CCX gates. However, for circuits requiring many sequential CCX operations on the same logical qubits, such as cascaded adders in Shor’s algorithm [22], the GSC approach eliminates the per-gate distillation cost entirely, and the relative overhead of the extension/splitting protocol is divided across multiple gates. Whether this trade-off favors GSCs depends on the target circuit depth, the physical error rate, and the available qubit budget.

B. Comparison with Concatenated Code-Switching

An alternative distillation-free route to universality uses concatenated quantum codes with complementary transversal gate sets. Jochym-O’Connor and Laflamme [23] demonstrated that concatenating the $[[7, 1, 3]]$ Steane code (which admits transversal Clifford gates) with the $[[15, 1, 3]]$ Reed–Muller code (which admits a transversal T gate) produces a $[[105, 1, 9]]$ code whose transversal gate set generates the full Clifford+ T group. Since $CNOT$ is transversal in both constituent codes, logical two-qubit gates remain transversal at every concatenation level. This construction achieves universal fault-tolerant quantum computation without magic state distillation or any special ancilla preparation beyond computational basis states.

The structural analogy to the GSC framework is that both approaches exploit complementary transversal gate sets by changing the encoding structure: in the concatenated approach, the change is between concatenated stabilizer-code layers; in the GSC approach, the change is between group labels on the edges of a two-dimensional topological patch. The mechanisms differ in several important respects, however, and the analogy should not be pushed too far.

In the concatenated construction, logical information is encoded jointly in an outer Steane code and an inner Reed–Muller code, both of which are qubit stabilizer codes with well-understood decoders. The cost of the scheme is the fixed encoding overhead (105 physical qubits per logical qubit at concatenation level 1) and the gate complexity of the inter-level encoding and decoding circuits. Accuracy is traded against overhead by increasing the concatenation level, which adds additional layers of encoding and error suppression.

In the GSC approach, logical information is transferred between $\mathbb{Z}_2$ surface code patches and a G -GSC patch via the extension/splitting protocol. The cost is the enlarged edge dimension ($\log_2 |G|$ qubits per edge) and the time depth of the extension/splitting rounds (d rounds per

transition). Accuracy is traded against overhead by increasing the code distance d , which increases the number of edges in the lattice. Crucially, the two-dimensional local geometry of the lattice is preserved throughout the protocol, whereas the concatenated construction does not natively preserve nearest-neighbor planar locality without additional routing or layout overhead.

1. Threshold and overhead comparison

Chamberland, Jochym-O’Connor, and Laflamme [24] established a depolarizing threshold of 1.28×10^{-3} for the $[[105, 1, 9]]$ concatenated code, and 9.69×10^{-4} for the smaller $[[49, 1, 5]]$ variant. However, a subsequent overhead analysis [25] found that the $\mathbb{Z}_2$ surface code supplemented with magic state distillation achieves a smaller space-time overhead than the concatenated scheme at the sampled physical error rates and noise models considered. An independent study by Beverland, Kubica, and Svore [26] reached a similar conclusion for two-dimensional/three-dimensional color code switching, reporting a code-switching T -gate threshold of only 0.07% under circuit-level noise—nearly an order of magnitude below the distillation threshold for the same architecture. This result pertains to color-code switching rather than GSC switching and should not be read as a threshold estimate for non-Abelian GSCs; it is included as evidence that code-switching overheads can be substantial even when the non-Clifford gate is transversal in the intermediate code.

For the GSC approach, no rigorous circuit-level threshold has been established for general non-Abelian groups. The single-fault correctness proven in this work (Theorem V.6) is a necessary condition for the existence of a threshold but does not imply one. For the special case of the quantum double of D_4 —a small non-Abelian example admitting a Toffoli-type transversal operation, with $|D_4| = 8$ and $q = 3$ qubits per edge—Jing et al. [11] have demonstrated non-Abelian charge-noise thresholds under perfect syndrome measurements: $p_c \approx 0.159$ for standard MWPM, $p_c \approx 0.208$ for intrinsically heralded MWPM, and $p_c \approx 0.218$ with an optimal decoder. These results suggest that GSC thresholds for small non-Abelian groups can approach surface-code levels, but the extension to circuit-level noise with imperfect syndromes and the full extension/splitting protocol remains open.

2. Architectural trade-offs

The three approaches—magic state distillation, concatenated code-switching, and GSC code-switching—occupy distinct points in the overhead landscape:

Magic state distillation has the most mature decoders and the highest demonstrated thresholds, with overhead scaling as $O(\log^\gamma(1/\epsilon))$ per non-Clifford gate for fixed gate arity [12]. Its disadvantage is the per-gate distilla-

tion cost, which is not amortized across sequential gates on the same qubits.

Concatenated code-switching eliminates distillation entirely and has a proven threshold [24], but does not natively preserve nearest-neighbor planar locality and trades accuracy for overhead through discrete concatenation levels rather than a continuous distance parameter.

GSC code-switching also eliminates distillation and preserves the two-dimensional local geometry of the surface code, with overhead adjustable via the code distance d . Its disadvantage is the exponential dependence of the edge dimension on gate arity (Sec. IV F) and the current absence of a proven circuit-level threshold for general non-Abelian groups.

For the full adder studied in this work, the GSC approach is most naturally compared with magic state distillation, since both operate within a two-dimensional surface code architecture. The concatenated approach offers a complementary alternative for platforms with non-planar connectivity, but the cited overhead and threshold studies suggest that it is less competitive in the planar-hardware regimes usually targeted by surface-code architectures.

C. Limitations

Several limitations of this work should be noted. The single-fault correctness result, Theorem V.6, establishes that the decoder corrects any single circuit-level fault, but does not establish a multi-fault threshold. The numerical simulations in Sec. VI operate on an effective class-chain noise model for the flux sector and do not simulate the charge sector, extension/splitting interfaces, or lattice surgery. The two-edge diagnostic reveals that, under a uniform enumeration of ordered nontrivial group-element pairs on adjacent edges, group-valued inter-class coupling causes 98.4% of tested local two-fault configurations to produce syndromes different from the per-edge class-chain syndrome, and 28.1% of these uniformly weighted configurations lead to logical failure when decoded classwise. These percentages are diagnostic combinatorial rates, not physical logical failure probabilities; an operational rate would require specifying the physical error channel on the encoded edge register and the syndrome-extraction circuit.

Additionally, the effective distance of the code during the column by column extension protocol requires careful analysis. While single faults produce localized defects correctable at any stage, a complete circuit-level threshold analysis accounting for time-dependent code geometry during extension is beyond the scope of this work.

D. Outlook

This work demonstrates that group surface codes can implement concrete quantum arithmetic circuits with an-

alytically proven single-fault correctness, using groups of modest size. Several directions for future work emerge naturally:

First, the two-edge diagnostic identifies a concrete target: a group-valued MWPM decoder that matches on exact face flux products rather than per edge class assignments. Such a decoder would close the gap between the classwise approximation and the full non-Abelian physics.

Second, extending Construction B to larger arithmetic circuits, such as modular exponentiation for Shor's algorithm, would test whether the GSC advantage scales favorably. The modular architecture is well suited to this: additional G_{CCX} patches can be introduced as needed, with the $\mathbb{Z}_2$ phases providing a low overhead idle mode between operations.

Finally, a circuit-level threshold analysis of the full Construction B protocol with realistic noise would determine whether the approach is competitive with magic state distillation at practical error rates.

Appendix A: G_{FA} Construction Derivations

a. Closure

We justify why the closure produces exactly the gates that generate H, K_1, K_2 . The generators CX_{ij} appear because products of the circuit gates yield all CNOT gates with control $i < j$ acting on target j . For instance, $CCX_{124} \cdot X_2 \cdot CCX_{124} = X_2 \cdot CX_{24}$, producing CNOT gates not originally in the circuit. The restriction to target qubit 4 for the CCX gates arises because the only non-Clifford generators are CCX_{124} and CCX_{234} , and their products with Clifford elements generate CCX_{134} , but no CCX gates targeting qubits 1, 2, or 3.

b. Verification of the Semi-direct Product Structure and Knit Product Structure

First we must show that H is normal in $H \rtimes K_1$. Conjugation of elements of H by K_1 generators must yield elements of H . As a representative example, $CX_{12}X_1CX_{12} = X_1X_2$, since $CX_{12}|a, b, c, d\rangle = |a, a \oplus b, c, d\rangle$, so applying this to $|a, b, c, d\rangle$ gives $|a \oplus 1, a \oplus a \oplus b \oplus 1, c, d\rangle = |a \oplus 1, b \oplus 1, c, d\rangle$, which is exactly X_1X_2 . More generally, conjugating X_i by CX_{jk} flips qubit i and additionally flips qubit k if $i = j$. In all cases, this results in a product of X operators, and hence an element of H .

Next we consider $H \rtimes K_1$ in relation to K_2 . a counterexample shows that it is not normal:

$$CCX_{234}CX_{12}CCX_{234} = CX_{12} \cdot CCX_{134} \quad (A1)$$

Despite the absence of a second semi-direct product, every element of G_{FA} admits a unique normal-form factorization $g = h \cdot k_1 \cdot k_2$ with $h \in H$, $k_1 \in K_1$, $k_2 \in K_2$.

This follows from the order count $|H| \cdot |K_1| \cdot |K_2| = 16 \cdot 64 \cdot 8 = 2^{13} = |G_{FA}|$ together with the fact that H, K_1, K_2 generate the full group. Now let $S = \{hk_1k_2 : h \in H, k_1 \in K_1, k_2 \in K_2\}$, which has cardinality 8192, by enumeration. Thus this gives us a bijection, and the factorization defines the 13 binary normal-form coordinates $(\alpha, \beta, \gamma, \delta, a, b, c, d, e, f, p, q, r)$ used throughout the paper. This leads to the group structure $G_{FA} = (H \rtimes K_1) \rtimes K_2$.

c. Verification of the L_{FA} Logical Operator

Step 1: $L_{CCX_{124}}|g\rangle \rightarrow |CCX_{124} \cdot g\rangle$. Using the group multiplication rules:

$$(\alpha, \beta, \gamma, 0) \mapsto (\alpha, \beta, \gamma, \alpha\beta) \quad (\text{A2})$$

where the auxiliary coordinates undergo a data-dependent permutation that leaves the $|+\rangle$ state invariant, established in Sec. III A *Step 2*:

$$L_{CX_{12}}(\alpha, \beta, \gamma, \alpha\beta) \mapsto (\alpha, \alpha \oplus \beta, \gamma, \alpha\beta) \quad (\text{A3})$$

Step 3:

$$L_{CCX_{234}}(\alpha, \alpha \oplus \beta, \gamma, \alpha\beta) \mapsto (\alpha, \alpha \oplus \beta, \gamma, \alpha\beta \oplus (\alpha \oplus \beta)\gamma) \quad (\text{A4})$$

Step 4:

$$L_{CX_{23}}(\alpha, \alpha \oplus \beta, \gamma, \alpha\beta \oplus (\alpha \oplus \beta)\gamma) \quad (\text{A5})$$

$$\mapsto (\alpha, \alpha \oplus \beta, \alpha \oplus \beta \oplus \gamma, \alpha\beta \oplus (\alpha \oplus \beta)\gamma) \quad (\text{A6})$$

Step 5:

$$L_{CX_{12}}(\alpha, \alpha \oplus \beta, \alpha \oplus \beta \oplus \gamma, \alpha\beta \oplus (\alpha \oplus \beta)\gamma) \quad (\text{A7})$$

$$\mapsto (\alpha, \beta, \alpha \oplus \beta \oplus \gamma, \alpha\beta \oplus (\alpha \oplus \beta)\gamma) \quad (\text{A8})$$

which is the desired result.

Appendix B: Algebraic Properties of G_{CCX} and $H \rtimes K_1$

We justify the algebraic properties stated in Propositions V.2–V.5, which are essential to the classwise MWPM decoder. All results are verified by exhaustive computation over the relevant group elements, with the verification script provided in the supplementary material.

Reproducibility of computational algebra. The following quantities are computed programmatically for each group: the full multiplication table, the conjugacy class decomposition (by exhaustive conjugation), the inverse of every element and verification that each conjugacy class is closed under inversion, the class algebra structure constants, the character table (by simultaneous diagonalization of the class algebra matrices, verified against row

and column orthogonality), the irreducible representation dimensions and characters, and the fusion multiplicities $\langle R_i \otimes R_j, \mathbf{1} \rangle$ for all pairs of irreps. Where a proposition relies solely on exhaustive computation rather than an analytic proof, this is stated explicitly.

<https://github.com/Nickolas-Ellison1/gsc-full-adder>.

a. Conjugacy Classes of G_{CCX}

The group G_{CCX} has order 64, and decomposes as $(H \rtimes K_1) \rtimes K_2$, with normal form coordinates $(\alpha, \beta, \gamma, a, b, p) \in \{0, 1\}^6$, where the multiplication rule is given by:

$$\begin{aligned} &(\alpha_1, \beta_1, \gamma_1, a_1, b_1, p_1) \cdot (\alpha_2, \beta_2, \gamma_2, a_2, b_2, p_2) \\ &= (\alpha_1 \oplus \alpha_2, \beta_1 \oplus \beta_2, \gamma_1 \oplus \gamma_2 \oplus a_1\alpha_2 \oplus b_1\beta_2, \\ &\quad a_1 \oplus a_2 \oplus p_1\beta_2, b_1 \oplus b_2 \oplus p_1\alpha_2, p_1 \oplus p_2) \end{aligned} \quad (\text{B1})$$

Exhaustive conjugation over all 64 elements yields 16 conjugacy classes with size distribution $\{1^2, 2^3, 4^8, 8^3\}$, where x^y means there are y conjugacy classes of size x , listed with representatives in Table III

TABLE III. Conjugacy classes of G_{CCX} . Each representative is given in normal-form coordinates $(\alpha, \beta, \gamma, a, b, p)$.

Class	$ C $	Order	Representative
C_0	1	1	(0, 0, 0, 0, 0, 0)
C_1	1	2	(0, 0, 1, 0, 0, 0)
C_2	2	2	(0, 0, 0, 1, 0, 0)
C_3	2	2	(0, 0, 0, 0, 1, 0)
C_4	2	2	(0, 0, 0, 1, 1, 0)
C_5	4	2	(0, 0, 0, 0, 0, 1)
C_6	4	2	(0, 0, 0, 1, 1, 1)
C_7	4	2	(0, 1, 0, 0, 0, 0)
C_8	4	4	(0, 1, 0, 0, 1, 0)
C_9	4	2	(1, 0, 0, 0, 0, 0)
C_{10}	4	4	(1, 0, 0, 1, 0, 0)
C_{11}	4	2	(1, 1, 0, 0, 0, 0)
C_{12}	4	4	(1, 1, 0, 1, 0, 0)
C_{13}	8	4	(0, 1, 0, 0, 0, 1)
C_{14}	8	4	(1, 0, 0, 0, 0, 1)
C_{15}	8	4	(1, 1, 0, 0, 0, 1)

b. Self-inverse Conjugacy Classes

For each of the 16 conjugacy classes, we verify that $C^{-1} = \{g^{-1} : g \in C\} = C$. This is trivially true for elements of order ≤ 2 . The remaining conjugacy classes $C_8, C_{10}, C_{12}, C_{13}, C_{14}, C_{15}$ have elements of order 4, and their self-conjugacy is verified through exhaustive computation.

c. Character Table and Self-duality

The character table is computed via simultaneous diagonalization of the class algebra structure constant matrices and verified against both row and column orthogonality relations. The 16 irreducible representations have dimensions $8 \times 1, 6 \times 2, 2 \times 4$ satisfying $8 \cdot 1^2 + 6 \cdot 2^2 + 2 \cdot 4^2 = 64$. Self-duality, $R \cong R^*$, follows from the self-inverse property of the conjugacy classes: $\chi_{R^*}(g) = \overline{\chi_R(g)} = \chi_R(g^{-1})$, and since g^{-1} is always conjugate to g (Proposition V.2), we have $\chi_R(g^{-1}) = \chi_R(g)$.

d. Unique Vacuum Fusion Channel

For each irrep R , the multiplicity of the trivial representation in $R \otimes R$ is computed via:

$$\langle R \otimes R, \mathbf{1} \rangle = \frac{1}{|G|} \sum_C |C| \chi_R(C)^2 \quad (\text{B2})$$

Evaluating all 16 irreps using the character table yields $\langle R \otimes R, \mathbf{1} \rangle = 1$ in every case. For the one dimensional

irreps, this follows from the formula immediately. For the character values $\pm 2, \pm 4$, we exhaustively compute each case. Additionally, we verify computationally that $\langle R_i \otimes R_j, \mathbf{1} \rangle$ vanishes.

1. Properties of $H \rtimes K_1$

The group $H \rtimes K_1 = \langle X_1, X_2, X_3, CX_{13}, CX_{23} \rangle$ has order 32 and is a subgroup of G_{CCX} . By exhaustive computation over all 32 elements, we verify:

1. $H \rtimes K_1$ has 17 conjugacy classes that are self-inverse, consisting of one trivial class, one class $\{X_3\}$, and 15 order 2 classes.
2. There are 17 irreducible representations, and since all the conjugacy classes are self-inverse, the same argument as for G_{CCX} gives self-duality for all irreps. For every irrep, the fusion $R \otimes R$ contains the trivial representation with multiplicity exactly 1, verified by direct computation as for G_{CCX} .

These properties ensure that the classwise MWPM flux and charge decoders apply to $H \rtimes K_1$.

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